

EVERY TAXICAB DISTANCE ONCE

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ABSTRACT. Can n points of the integer grid be placed so that their $\binom{n}{2}$ taxicab distances are $1, 2, \dots, \binom{n}{2}$, each exactly once? On a line this asks for a perfect Golomb ruler, which exists only for $n \leq 4$. A parity count shows that n or $n - 2$ must be a perfect square, and placements were known for $n = 2, 3, 4, 6, 9$. We give a placement of eleven points, the first case beyond those, so a placement exists for every $n \leq 15$ that the parity count allows. The parity count and the eleven-point placement are formally verified in Lean.

1. THE QUESTION

Elliott asked on Mathematics Stack Exchange in September 2024 [1]: for which n can n points be placed on the integer grid so that, in the taxicab distance $|x_1 - x_2| + |y_1 - y_2|$, the $N = \binom{n}{2}$ pairs are at distances $1, 2, \dots, N$, each exactly once? For $n = 3$ the points $(0, 0)$, $(0, 2)$, $(0, 3)$ work, with distances 2, 3 and 1. Before reading on, try $n = 4$, and then $n = 5$.

On a single line the question asks for a perfect Golomb ruler: marks at which every length from 1 to the total length is measured exactly once. Equivalently, it asks for a graceful labelling of the complete graph K_n , and Golomb showed that one exists only for $n \leq 4$ [2]. The grid gives the points a second direction, and the answer changes: the asker found placements for $n = 6$ and $n = 9$ by computer, proved that n or $n - 2$ must be a perfect square, and noted that the program was too slow to go further. Its only answer is the author's, giving the placement below.

Theorem 1. *If n points realise every distance $1, \dots, N$ once, and a of them have $x + y$ even and b have $x + y$ odd, then*

$$(a - b)^2 = \begin{cases} n, & n \equiv 0, 1 \pmod{4}, \\ n - 2, & n \equiv 2, 3 \pmod{4}. \end{cases}$$

In particular n is a perfect square when $n \equiv 0, 1 \pmod{4}$, and $n - 2$ is a perfect square otherwise.

Proposition 2. *The eleven points*

$$\begin{aligned} &A(0, 14), B(0, 16), C(1, 16), D(7, 11), E(8, 7), F(11, 2), \\ &G(12, 7), H(14, 45), I(17, 49), J(23, 0), K(23, 26) \end{aligned}$$

have taxicab distances $1, 2, \dots, 55$, each exactly once.

The values of n allowed by Theorem 1 begin 2, 3, 4, 6, 9, 11, 16, 18, 25, 27, $\dots$. So a placement exists for $n = 2, 3, 4, 6, 9, 11$ and for no other $n \leq 15$, and the question is open from $n = 16$ on.

2. THE PARITY COUNT

Colour each grid point by the parity of $x + y$. A taxicab distance $|x_1 - x_2| + |y_1 - y_2|$ has the parity of $(x_1 + y_1) - (x_2 + y_2)$, so it is odd exactly when its endpoints have different colours.

Proof of Theorem 1. The pairs of different colours number ab , and they realise exactly the odd distances. Among $1, \dots, N$ there are $\lceil N/2 \rceil$ odd numbers, so $ab = \lceil N/2 \rceil$, and

$$(a - b)^2 = (a + b)^2 - 4ab = n^2 - 2N - 2(N \bmod 2) = n - 2(N \bmod 2),$$

using $2N = n^2 - n$. Finally $N = n(n - 1)/2$ is even exactly when 4 divides $n(n - 1)$, that is, when $n \equiv 0$ or $1 \pmod{4}$. $\square$

For $n = 5$ the count asks for $(a - b)^2 = 5$, which is impossible; that is the answer to the warm-up. The proof also fixes the colour classes: their sizes are $\frac{1}{2}(n \pm \sqrt{n})$ or $\frac{1}{2}(n \pm \sqrt{n-2})$. For $n = 11$ that is seven points of one colour and four of the other.

3. ELEVEN POINTS

Figure 1 shows the placement, and Table 1 proves Proposition 2. Each entry of its upper half is one subtraction in each coordinate and an addition: the distance from H to J, for instance, is $|23 - 14| + |0 - 45| = 54$. The lower half names, for each d from 1 to 55, a pair that the upper half shows is at distance d . Fifty-five different distances need fifty-five different pairs, and there are only fifty-five pairs, so each is used once. As Theorem 1 requires, seven points (C, E, F, G, H, J, K) have $x + y$ odd and four have it even, and $(7 - 4)^2 = 11 - 2$.

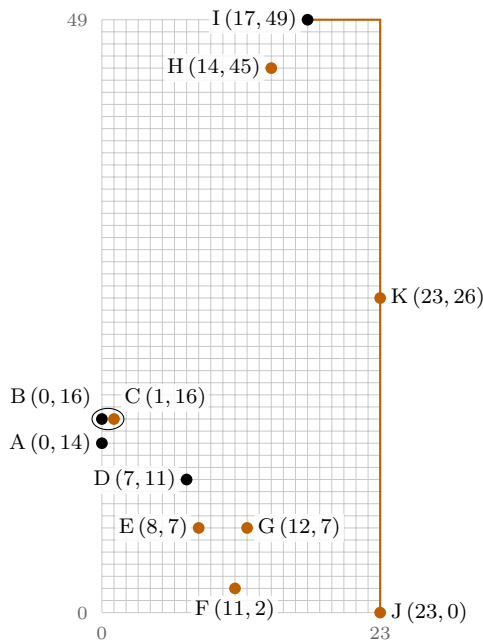


Figure 1. The eleven points of Proposition 2 in the box $0 \leq x \leq 23$, $0 \leq y \leq 49$, coloured by the parity of $x + y$: seven copper (odd), four black (even). Every odd distance joins a copper point to a black one. The ringed pair B, C is at distance 1; the path from I to J, 6 steps across and 49 down, has length 55.

The smaller placements show how the second dimension helps (Figure 2). For $n = 3, 4$ the points lie on one line, as the Golomb rulers $\{0, 2, 3\}$ and $\{0, 2, 5, 6\}$. For $n = 6$ two adjacent lines suffice. For $n = 9$ and $n = 11$ the placements found by computer are scattered across the plane.

4. HOW THE PLACEMENT WAS FOUND, AND WHAT IS OPEN

An exact search places the distances from the largest down: the largest distance not yet realised must join a new point to a placed one, or two new points, and every branch is tried. It finds the placements for $n \leq 6$, but its running time grows by a factor of about 35 with each extra point, so it cannot reach $n = 11$. The eleven points came from simulated annealing instead, moving one point at a time and accepting moves that reduce, or occasionally increase, the number of distances in $1, \dots, N$ not yet realised. It finds placements for $n = 6$ and 9 in under a second and for $n = 11$ in under half a minute.

For $n = 16$, the next value the parity count allows, a run of 100 seconds found nothing. That reflects a short run and is no evidence that sixteen points are impossible. Whether every n allowed by Theorem 1 can be realised is open; neither the question nor its comments offers any obstruction beyond parity.

	A	B	C	D	E	F	G	H	I	J
B	2									
C	3	1								
D	10	12	11							
E	15	17	16	5						
F	23	25	24	13	8					
G	19	21	20	9	4	6				
H	45	43	42	41	44	46	40			
I	52	50	49	48	51	53	47	7		
J	37	39	38	27	22	14	18	54	55	
K	35	33	32	31	34	36	30	28	29	26

1	BC	12	BD	23	AF	34	EK	45	AH
2	AB	13	DF	24	CF	35	AK	46	FH
3	AC	14	FJ	25	BF	36	FK	47	GI
4	EG	15	AE	26	JK	37	AJ	48	DI
5	DE	16	CE	27	DJ	38	CJ	49	CI
6	FG	17	BE	28	HK	39	BJ	50	BI
7	HI	18	GJ	29	IK	40	GH	51	EI
8	EF	19	AG	30	GK	41	DH	52	AI
9	DG	20	CG	31	DK	42	CH	53	FI
10	AD	21	BG	32	CK	43	BH	54	HJ
11	CD	22	EJ	33	BK	44	EH	55	IJ

Table 1. Top: the distance between the points labelling each row and column. Bottom: the same 55 numbers sorted, each with the pair that realises it; reading down the columns runs through $1, \dots, 55$.

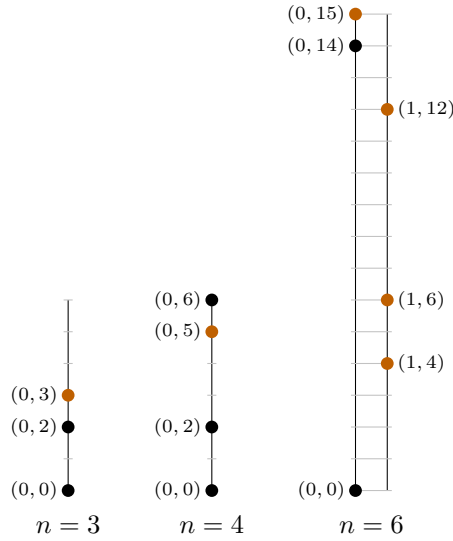


Figure 2. Placements for $n = 3, 4$ and 6 , coloured by the parity of $x + y$. The first two are Golomb rulers; the six points $(0, 0)$, $(0, 14)$, $(0, 15)$, $(1, 4)$, $(1, 6)$, $(1, 12)$ use a second line, one step away.

5. VERIFICATION

Theorem 1 and Proposition 2, including the colour split, are proved in Lean 4 with Mathlib [3]: the statement is that the distances of the pairs $i < j$ form the set $\{1, \dots, N\}$ and are pairwise different, and the eleven-point case is a finite check. A Python script checks the parity rule for $n \leq 400$ and every placement above, and catches two planted errors (counting $\lfloor N/2 \rfloor$ odd numbers, and moving one point). Code and Lean files are at [4].

ACKNOWLEDGMENTS

The question, the parity condition and the first placements, for $n \leq 9$, are Elliott's (the placements shown here come from our own searches); the remark on Golomb rulers is a comment by Alex Ravsky, and the link with graceful labellings was noted in a comment by vallev. The search, the analysis and the Lean formalisation were carried out with Claude Opus 5.5 (Anthropic). The author is responsible for the content.

REFERENCES

- [1] Elliott, *The existence of patterns with unique pairwise distances*, Mathematics Stack Exchange question 4967838 (2024), <https://math.stackexchange.com/q/4967838>; the eleven-point placement was posted as answer 5150779 (2026).
- [2] S. W. Golomb, *How to number a graph*, in: Graph Theory and Computing, ed. R. C. Read, Academic Press, New York, 1972, 23–37.
- [3] The mathlib Community, *The Lean mathematical library*, in: Proc. 9th ACM SIGPLAN Int. Conf. on Certified Programs and Proofs (CPP 2020), 367–381.
- [4] V. Jandhyala, code and Lean proofs, <https://github.com/jvvk/mathematics/tree/main/every-taxicab-distance-once>.

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